

Weekly Report

Qlab-Oriented SDCS Small-Map Planning and 2D Vehicle-Following Controller Design

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This week has two connected parts:

- **SDCS small-map planning baseline:** transferred the Qlab-oriented roadmap and smooth-turn approach into the current QCar/CARLA project, then checked the resulting route in CARLA.
- **2D vehicle-following controller design:** formulated the Chapter 3 look-ahead controller for the QCar bicycle model and identified the remaining observer and steering constraints.

The CARLA result in this report is a path-planning and path-following baseline. The 2D platoon controller and distributed high-gain observer are still design work and are not claimed as validated results.

2 SDCS planner: Qlab-to-project baseline

2.1 Reference and transfer objective

The SDCS small-map planner was oriented from the Qlab path-planning reference. The reference provides the roadmap and smooth-turn idea; the project version brings this idea into the current QCar/CARLA environment as a self-contained planner for the SDCS small map.

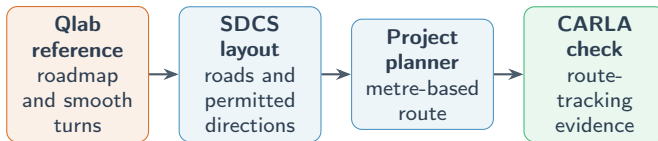


Figure 1: Transfer path from the Qlab planning reference to the project-local SDCS small-map baseline.

2.2 What was retained and adapted

Aspect	Qlab reference	Current project planner
Purpose	General roadmap support for Quanser environments.	Route generation for the SDCS small map used by QCar/CARLA.
Road layout	A reusable map structure made of locations and road connections.	Eleven fixed small-map locations with permitted directed connections.
Route choice	A valid shortest connection between requested locations.	A valid right-hand-traffic route through requested small-map locations.
Path shape	Straight sections and smooth turns.	The same route-shape idea, sampled as tracking waypoints.
Execution environment	Uses the Qlab support environment.	Runs inside this project without requiring the licensed Qlab library at runtime.
Evidence	Reference implementation.	CARLA route-tracking artifact in Figure 3.

Table 1: Comparison between the Qlab reference and the project-local SDCS planner.

2.3 Planner outcome and current integration

The planner receives a requested sequence of map locations, connects them through permitted roads, and returns a smooth waypoint route. A route can end at its requested destination or repeat as a closed circuit. This gives the control system a repeatable path reference for CARLA tests.

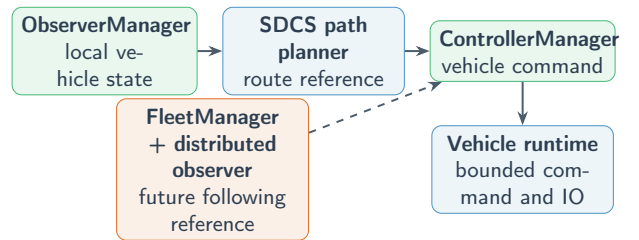


Figure 2: Current SDCS path-reference flow and the future fleet-following integration point.

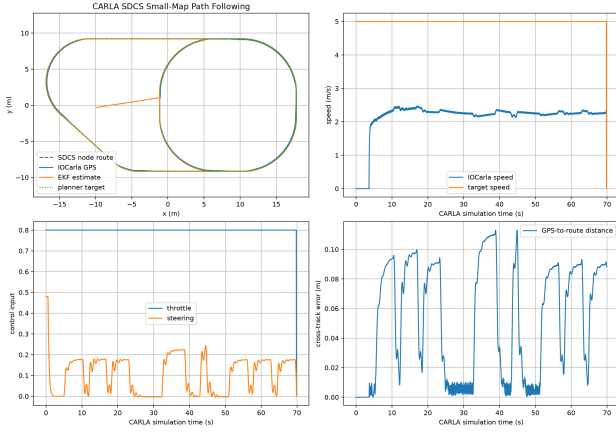


Figure 3: Quanser SDCS map trajectory following under CARLA with ego vehicle. The planner target, estimated path, speed, commands, and cross-track error are recorded for review.

3 2D following-control formulation

3.1 Reference model and time-gap objective

The following formulation is based on Chapter 3 of Bayuwindra's PhD thesis on look-ahead tracking controllers for integrated longitudinal and lateral vehicle-platoon control.¹

For vehicle i , the reference model is

$$\dot{x}_i = v_i \cos \psi_i, \quad \dot{y}_i = v_i \sin \psi_i, \quad \dot{v}_i = a_i, \quad \dot{\psi}_i = \omega_i, \quad (1)$$

with

$$u_i^{\text{ref}} = [a_i \quad \omega_i]^\top. \quad (2)$$

$$p_i = [x_i \quad y_i]^\top, \quad (3)$$

$$d_{r,i} = (r_i + h_i v_i) \begin{bmatrix} \cos \psi_i \\ \sin \psi_i \end{bmatrix}, \quad (4)$$

$$e_i = (p_{i-1} - p_i) - d_{r,i}. \quad (5)$$

3.2 Error-state feedback form

Error Dynamics:

$$z_i = \begin{bmatrix} e_{x,i} \\ e_{y,i} \\ v_{i-1} \cos \psi_{i-1} - v_i \cos \psi_i \\ v_{i-1} \sin \psi_{i-1} - v_i \sin \psi_i \end{bmatrix}. \quad (6)$$

Control input:

$$\begin{bmatrix} a_i \\ \omega_i \end{bmatrix} = F_i^{-1} \begin{bmatrix} z_{3,i} + k_{1,i} z_{1,i} \\ z_{4,i} + k_{2,i} z_{2,i} \end{bmatrix}, \quad (7)$$

¹A. Bayuwindra, *Look-ahead tracking controllers for integrated longitudinal and lateral control of vehicle platoons*, PhD thesis, Eindhoven University of Technology, 2019, Chapter 3.

where $k_{1,i}$ and $k_{2,i}$ are controller parameters, and

$$F_i = \begin{bmatrix} h_i \cos \psi_i & -(r_i + h_i v_i) \sin \psi_i \\ h_i \sin \psi_i & (r_i + h_i v_i) \cos \psi_i \end{bmatrix}. \quad (8)$$

Remark: Note that the control input is for 2 wheel unicycle vehicles. So the yaw rate ω_i is directly commanded, which is not the case for 4 wheel vehicles. The bicycle model transformation is needed to convert the yaw-rate command into a front-wheel steering command.

3.3 Bicycle-model transformation

For 4 wheel vehicle i is a bicycle model with wheelbase L and front-wheel steering $\delta_{f,i}$. The bicycle model is transformed into the unicycle form in (1) by

$$\dot{\psi}_i = \frac{v_i}{L} \tan \delta_{f,i}, \quad \kappa_i = \frac{1}{L} \tan \delta_{f,i}, \quad (9)$$

$$\omega_i = v_i \kappa_i.$$

$$\begin{bmatrix} \ddot{x}_i \\ \ddot{y}_i \end{bmatrix} = \underbrace{\begin{bmatrix} \cos \psi_i & -v_i^2 \sin \psi_i \\ \sin \psi_i & v_i^2 \cos \psi_i \end{bmatrix}}_{H_i} \begin{bmatrix} a_i \\ \kappa_i \end{bmatrix}. \quad (10)$$

$$F_i^{\text{bike}} = \begin{bmatrix} h_i \cos \psi_i & -v_i(r_i + h_i v_i) \sin \psi_i \\ h_i \sin \psi_i & v_i(r_i + h_i v_i) \cos \psi_i \end{bmatrix}, \quad (11)$$

$$\begin{bmatrix} a_i \\ \kappa_i \end{bmatrix} = (F_i^{\text{bike}})^{-1} \begin{bmatrix} z_{3,i} + k_{1,i} z_{1,i} \\ z_{4,i} + k_{2,i} z_{2,i} \end{bmatrix}. \quad (12)$$

$$\delta_{f,i} = \arctan(L \kappa_i). \quad (13)$$

3.4 Next step: Observable-form target and constraints

- Triangular-form representation of controlled platoon model:

$$\zeta_i = T(q_i) = [X_i \quad \dot{X}_i \quad \ddot{X}_i \quad Y_i \quad \dot{Y}_i \quad \ddot{Y}_i]^\top. \quad (14)$$

$$\dot{\zeta}_{i,j} = \zeta_{i,j+1}, \quad (15)$$

$$\dot{\zeta}_{i,r} = \varphi_i(\zeta_i, u_i) + d_i, \quad y_i = C \zeta_i.$$

$$\boxed{T(\cdot), \varphi_i(\cdot), \text{ and } y_i \text{ remain to be derived.}} \quad (16)$$

- The constraints on the bicycle model: steering angle $\delta_{f,i}$ is bounded, for qcar $\delta_{\max} = \frac{1}{6}\pi$ rad.

$$\boxed{\psi_i \notin u_i^{\text{bike}}, \quad \delta_{f,i} \in [-\delta_{\max}, \delta_{\max}], \quad \delta_{f,i}^* = \arctan\left(\frac{L \omega_i^*}{v_i}\right), \quad v_i > 0.} \quad (17)$$

- *Observer form:* derive $T(\cdot)$ and y_i for an observable canonical or triangular form.
- *Yaw and steering:* yaw is reached only indirectly through limited steering; the yaw-rate-to-steering conversion is unsuitable near zero speed.